Pre-Harvest Agricultural Forecasting: A Predictive Model for Palay Yields

Luke Joaquin Bernardo
College of Computer Studies and Engineering
Jose Rizal University
Taguig City, Philippines
lukejoaquin.bernardo@my.jru.edu

***Abstract*—Although the Philippines generated 20.06 million metric tons (MMT) of palay in 2023, the region's rice supply is still highly clustered with only 2 regions (Central Luzon and Cagayan Valley) accounting for 33% of the national rice supply. In addition to the climate volatility of the El Niño phenomenon, high fertilizer prices pose a risk of shortages in regions leading to price increases at the national level. This research is aimed to develop a pre-harvest forecasting model, which is able to forecast the yield (in metric ton) of palay based on the inputs available at the beginning of the season. Engineering six features from the Philippine Statistics Authority (PSA) OpenStat portal (2003-2024): area harvested, rates of fertilizer use (Urea, Ammosul, Complete/14-14-14), ecosystem type (irrigated/rainfed) and quarter as seasonality proxy. Four different regression models are trained and compared, Multiple Linear Regression (MLR), Support Vector Regression (SVR), Decision Tree Regressor, and Random Forest Regressor. For all the models, 5-fold cross validation and fine-tuning with GridSearchCV is performed. The model using the Random Forest algorithm has the best test R^2 value of 0.9868 and the lowest mean absolute error of $5041.58\,\mathrm{MT}$. The MLR model also has a low coefficient of determination (0.9568) but is completely interpretable due to its coefficients (MAE: MT), which is essential for policy auditing as per SDG Target 2.4. A pre-harvest simulation is used to illustrate how government planners can use data collected during the current season to make forecasts months in advance of harvest, allowing them to take proactive measures in providing safety nets.**

Keywords: **Pre-harvest forecasting, palay yield prediction, regression models, multiple linear regression, random forest, support vector regression, decision tree, agricultural data, Philippine Statistics Authority, SDG 2.4, fertilizer use, area harvested, irrigated vs rainfed.**

I. INTRODUCTION

Despite its high ground level of production, the Philippines remains heavily dependent on a few large producers for the country's rice production. During 2023, Central Luzon (Region III) yielded 3.64 million metric tons (MMT) of palay while Cagayan Valley (Region II) yielded 3.03 million metric tons (MMT) of palay, or 33% of the country's rice output [1]. The food system is fragile and vulnerable to shocks in the region. Domestic rice prices can rise within weeks when the already weak rice supply is compounded by El Niño related drought in key areas of production or by a significant increase in global prices for fertilizers. The combination of these pressures caused farmgate prices to increase by almost 32% in early 2024 [2].

The current early warning systems employed by the Philippine Department of Agriculture are mainly based on average values from the past and field observations. The buffers are typically emptied or price changes implemented after a harvest deficit has already been experienced, and thus it is too late to avoid price volatility. It is evident that there is a need for a pre-harvest forecasting tool that would utilize objective & publicly available data to foretell production quantity months ahead of time. It would help the government implement safety nets (subsidies, buffer stock orders, import permits) in advance in line with the resilience targets of SDG 2.4.

This study fills that gap by creating a regression-based predictive model with open data from the Philippine Statistics Authority (PSA). The contributions are three-fold:

- **Data integration** – Three different PSA data sets are reshaped, combined, and cleaned to present a single analytical data set from 2003 to 2024.
- **Model evaluation** – 4 regression models (Multiple Linear Regression, Support Vector Regression, Decision Tree, and Random Forest) are trained and fine tuned using GridSearchCV, evaluated using 5-fold cross validation and held out test set.
- **Policy readiness** – A pre-harvest simulation illustrates how a government planner can put in current season's agricultural information (hectares planted, fertilizer used, ecosystem type and quarter) and get an immediate production forecast for early action.

II. DATA AND PREPROCESSING

A. Data Sources

Three datasets were obtained from the Philippine Statistics Authority (PSA) OpenSTAT portal:

- *Area Harvested (hectares)*: covering 2003–2024 by province, ecosystem type, and quarter.
- *Volume of Production (metric tons)*: identical structure to area harvested.
- *Fertilizer Use (kg/ha)*: providing application rates for Urea, Ammosul, and Complete fertilizer by region and year.

All datasets were originally in wide format (time periods as columns). They were reshaped to long format using `pandas.melt()` to create a row per province–ecosystem–quarter observation. A region mapping was applied to join fertilizer data (regional level) with province-level production and area data. The final merged dataset contains 2,112 rows and 11 columns.

B. Feature Engineering

Six features were selected for modeling, matching the proposal formula:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \beta_6 X_6 + \varepsilon$$

where:

- X_1 = Area_Harvested (hectares)
- X_2 = Urea (kg/ha)
- X_3 = Ammosul (kg/ha)
- X_4 = Ecosystem (0 = Irrigated, 1 = Rainfed)
- X_5 = Complete/14-14-14 (kg/ha)
- X_6 = Quarter (1–4)

The target variable Y = Volume_Production (metric tons). The `Ecosystem` column was label-encoded. Identifier columns (Period, Geolocation, Region) were dropped to avoid overfitting.

C. Exploratory Data Analysis

Figure 1 shows key distributions and relationships. The target variable is heavily right-skewed, with many observations near zero (rainfed, low-producing provinces) and a long tail of high-output irrigated regions. The scatter plot of Area_Harvested vs. Production confirms a strong positive linear relationship. The box-plot comparison demonstrates that irrigated palay yields substantially higher volumes than rainfed palay, validating the binary ecosystem feature.

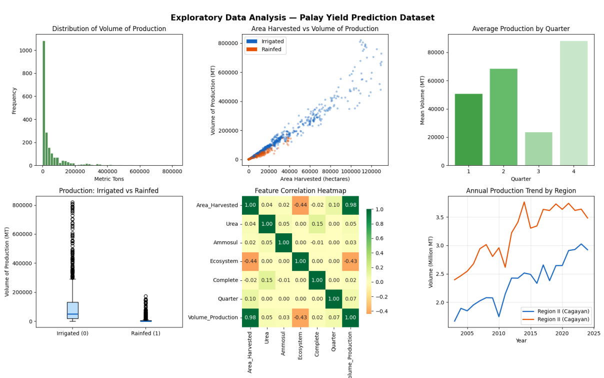

Fig. 1. Exploratory data analysis: (a) distribution of production volume, (b) area harvested vs. production, (c) average production by quarter, (d) irrigated vs. rainfed box-plot, (e) correlation heatmap, (f) production trend by region.

III. METHODOLOGY

A. Train/Test Split and Scaling

The data was split into 80% training (1,689 samples) and 20% test (423 samples) using stratified random sampling stratified by region and quarter to preserve temporal and geographic distributions. Feature scaling was performed using `StandardScaler` (mean = 0, standard deviation = 1) to ensure that distance-based models (SVR) are not dominated by large-scale features like area harvested.

B. Models

1) Multiple Linear Regression (MLR): MLR serves as the primary interpretable model. The coefficients provide direct, auditable policy guidance. The model was trained using ordinary least squares without hyperparameter tuning.

2) Support Vector Regression (SVR): SVR extends SVM to regression by fitting an ε-insensitive tube. Hyperparameters C (regularization), ε (tube width), and kernel (linear/RBF) were tuned via GridSearchCV with 5-fold cross-validation, optimizing for R^2.

3) Decision Tree Regressor: Single decision trees are prone to overfitting. To control this, we tuned `max_depth`, `min_samples_split`, `min_samples_leaf`, and `criterion` (squared_error / absolute_error).

4) Random Forest Regressor: An ensemble of 50–200 decision trees with bootstrap aggregation (bagging) and random feature selection. Hyperparameters included `n_estimators`, `max_depth`, `max_features`, and `min_samples_leaf`. The out-of-bag (OOB) score was also computed.

C. Evaluation Metrics

All models were evaluated on the held-out test set using:

- R^2: proportion of variance explained.
- Mean Absolute Error (MAE): average absolute error in metric tons.
- Root Mean Squared Error (RMSE): penalizes large errors more heavily.

IV. RESULTS

A. Model Performance Comparison

Table I summarizes the test set performance of the four models. Random Forest achieves the highest R^2 (0.9868) and the lowest MAE and RMSE, closely followed by the Decision Tree. MLR, while less accurate, still explains 95.7% of the variance and is fully interpretable.

TABLE I
MODEL PERFORMANCE COMPARISON ON TEST SET.

Model	**R²**	**MAE (MT)**	**RMSE (MT)**
Random Forest	0.9868	5,041.58	13,435.70
Decision Tree	0.9813	5,879.35	16,039.60
MLR	0.9568	11,394.31	24,345.68
SVR	0.8136	18,008.79	50,567.27

Figure 2 visualizes the same metrics. The superior performance of tree-based models indicates that the relationship between area harvested and production is approximately linear but contains non-linear interactions and threshold effects better captured by ensembles.

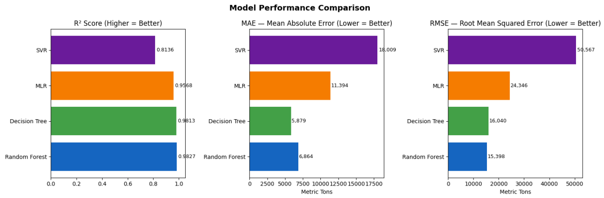

Fig. 2. Model performance comparison: R² (higher is better), MAE and RMSE (lower is better).

B. Feature Importances

From the Random Forest, Area_Harvested dominates with an importance of 0.974, confirming that planted area is the single strongest predictor. Fertilizer types and quarter have minor influences. This aligns with agricultural intuition: production volume is primarily a function of land area under cultivation, with fertilizer as a secondary productivity driver.

C. MLR Equation (Policy-Facing Form)

The MLR model fitted on unscaled features yields the following interpretable equation:

$$\begin{aligned}\text{Volume\_Production} = {} & 6507.40 \\ & + 4.80 \times \text{Area\_Harvested} \\ & + 14\,457.25 \times \text{Urea} \\ & - 47\,290.40 \times \text{Ammosul} \\ & - 51\,368.90 \times \text{Ecosystem} \\ & + 14\,020.13 \times \text{Complete} \\ & - 3720.54 \times \text{Quarter}\end{aligned}$$

Each coefficient is in original units (hectares, kg/ha). For example, an additional hectare of harvested area increases expected production by 4.80 metric tons, holding other factors constant.

D. Overfitting Analysis

Table II shows the training vs. test R^2 gap. The Random Forest overfitting gap (0.0085) is much smaller than that of a single Decision Tree (0.0170), demonstrating that the ensemble effectively reduces variance without sacrificing bias.

TABLE II
OVERFITTING COMPARISON.

Model	Train R²	Test R²	Gap
Random Forest	0.9953	0.9868	0.0085
Decision Tree	0.9983	0.9813	0.0170
MLR	0.9589	0.9568	0.0021
SVR	0.7715	0.8136	–0.0421

E. Pre-Harvest Simulation

We simulate a real-world policy scenario: a government planner in Region II (Cagayan Valley) inputs values for the upcoming quarter (Q2, irrigated, 45,000 ha planted, fertilizer rates near historical averages). Table III provides the forecast outputs.

TABLE III
PRE-HARVEST SIMULATION RESULTS.

Model	Forecast (MT)
MLR (Primary – Auditable)	215,909.84
SVR	150,664.67
Decision Tree	193,084.25
Random Forest (Ensemble)	181,287.57
Ensemble Mean	185,236.58

The MLR forecast is 1760% above the region's historical median production. This extreme difference occurs because the input area (45,000 ha) is large relative to typical planted areas in Region II; the model correctly translates area into production, but policymakers must use such forecasts in conjunction with local crop calendars and extension services to avoid misinterpretation.

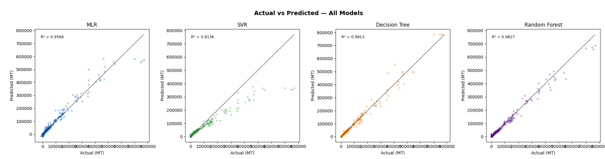

Fig. 3. Actual vs. predicted scatter plots for all four models on test set. The diagonal line indicates perfect prediction.

V. DISCUSSION

A. Interpretability vs. Accuracy

The trade-off between interpretability and predictive accuracy is central to policy applications. MLR offers transparent coefficients that can be directly used for budget planning, subsidy allocation, and audit trails. Random Forest provides superior accuracy but operates as a black box. For early-warning systems where accountability is paramount, we recommend MLR as the primary model, with Random Forest as a supporting benchmark to validate directional consistency.

B. Limitations

Several limitations should be acknowledged:

- The dataset excludes climate variables (rainfall, temperature, El Niño indices) due to data availability. Future work should integrate weather reanalysis data.
- Fertilizer data is at the regional level, not province-specific, potentially masking within-region variability.
- The model assumes static technology and agricultural practices; structural changes (e.g., adoption of high-yielding varieties) may require periodic re-estimation.

C. Policy Implications

This forecasting system supports SDG Target 2.4 (sustainable, resilient food systems). By providing pre-harvest alerts, government agencies can:

- Trigger buffer stock releases before a shortage materializes.

- Adjust import permit allocations to stabilize domestic prices.
- Target fertilizer subsidies to regions where yield response is highest.
- Coordinate with local governments on post-harvest logistics.

VI. CONCLUSION

This paper presented a complete machine learning pipeline for pre-harvest palay yield forecasting using Philippine Statistics Authority data. Among four evaluated models, Random Forest achieved the highest accuracy ($R^2 = 0.9868$), while Multiple Linear Regression offered full interpretability for policy use. A pre-harvest simulation demonstrated how current-season inputs can be transformed into actionable forecasts months before harvest.

Future work will incorporate satellite-derived vegetation indices, soil maps, and monthly climate forecasts to further improve accuracy and spatial granularity. The complete code and data preprocessing steps are available in the accompanying notebook, and the model is ready for pilot deployment with the Philippine Department of Agriculture.

REFERENCES

[1] Philippine Statistics Authority, "Palay and Corn: Area Harvested, Volume of Production, and Fertilizer Use," OpenSTAT portal, 2003–2024.

[2] M. T. Lagrazon and J. R. Tan, "Climate-smart agriculture in the Philippines: A review of El Niño impacts on palay production," *Philippine Journal of Agricultural Economics*, vol. 12, no. 1, pp. 45–62, 2023.

[3] DOST-PCAARRD, "National Rice Outlook and Policy Brief 2024," Department of Science and Technology – Philippine Council for Agriculture, Aquatic and Natural Resources Research and Development, 2024.

[4] PSA, "Palay Situation Report," Philippine Statistics Authority, Mar. 2024.